Supplementary Material
SENSO: Ontology-Based Checking of Coding-Agent Manifests Against Repository History

This document holds detailed tables and analyses that support the main paper. References to main-paper sections use the prefix "main".

S1. Details of the SEO Ontology and the Reasoning Pipeline

This section gives the design details of SEO and SENSO that the main paper summarises in Section 3. It covers the modules, the boundary rules, the rule encodings, the threshold register, the claim module, the reasoners, the reasoning cost, and the freeze points and model panel of the evaluation protocol (Section 4.2.1).

S1.1. Design Method and Modules

SEO re-engineers an earlier version of the ontology. We kept the earlier entity and process vocabulary, whose scope had been set with competency questions in the manner of the NeOn methodology [1], and rebuilt every label as a rule. Those competency questions use the earlier property vocabulary and were not re-executed against the current modules, so they are not reported as evidence. We did not perform a formal metaproperty analysis of the class hierarchy.

Figure S1 shows the modules. Five vocabulary modules (*seo-entities*, *seo-processes*, *seo-metrics*, *seo-laws* and *seo-patterns*) define projects, commits, modules, manifests, evolution activities, the measured data properties, one individual per Lehman law and the anti-pattern concepts. Three rule modules hold every axiom that can assign a label. *seo-evidence* holds the commit-level rules, *seo-inference* the repository-level and module-level rules, and *seo-claims* the rules that give claim check results. A ninth module, *seo-legacy*, re-encodes the earlier label logic for regression testing only. The module *seo-claims* imports *seo-entities* only, so it cannot change any repository label.

Every numeric measured data property is functional and carries a unit and an annotation naming the Python function that computes it. Each label class carries an explanation template, for example "no commit for {*monthsSinceLastCommit*} months", which is filled from the facts of a justification.

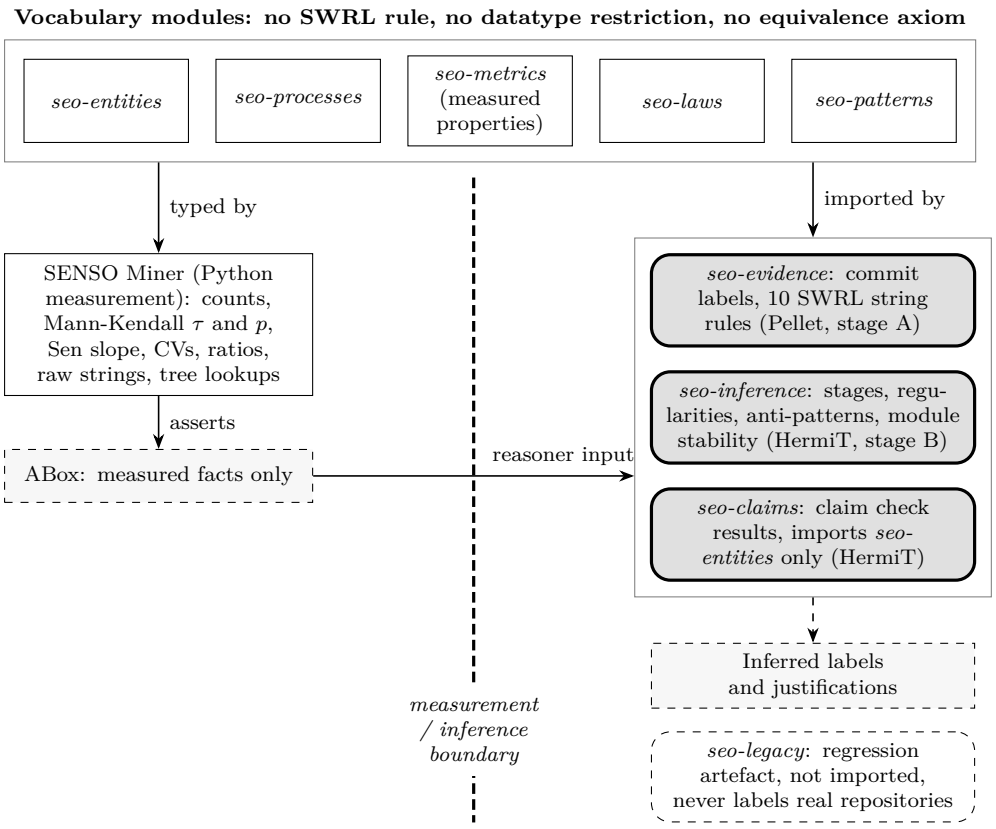


Figure S1: SEO Modules and the Measurement/Inference Boundary

S1.2. Rules of the Measurement/Inference Boundary

The SENSO Miner, the Python measurement component, may count, sum and average, and it may compute statistics that contain no cut-off, such as the Mann-Kendall statistics τ and p [2, 3], the Sen slope [4], coefficients of variation (CVs) and ratios of two measurements. It may extract raw strings such as co-author trailers, look up a normalised string in the file tree or in a parsed dependency manifest, and turn a version constraint into numeric bounds. The Miner may not compare a measurement with a constant, choose a branch based on a measurement, emit a boolean that encodes a threshold, or assert a label type. The ABox writer enforces the third rule by raising an error on boolean literals.

Each else-branch is a positive complement over a functional property (for example $x \geq k$ for "not $x < k$"), because the absence of a fact proves nothing under the open-world assumption. Comparisons of two measurements are expressed as a measured ratio compared with a constant. Repository measurements that depend on commit labels are aggregated between two reasoning stages without any threshold.

The boundary can be checked automatically. If the rule modules are removed and the reasoner is run again, no label should remain. Any label that survives must have been written by Python, which would break the boundary. For repository labels the check is structural, because the five vocabulary modules contain no SWRL rule, no datatype restriction and no equivalence axiom. For claim check results the check is executed. The self-test of the claims module reasons over 47 synthetic claims that cover every leaf class and the exact boundary values, with zero failures, and reasoning without *seo-claims* yields zero check results.

S1.3. Rule Encodings

C1 applies the two-sided Mann-Kendall test to the monthly LOC series (non-blank source lines at the last first-parent commit of each month). *GrowthStage* and *DeclineStage* require $p < \alpha$ with $\tau > 0$ and $\tau < 0$ respectively, and *StabilizationStage* requires $p \geq \alpha$. All three require a series of at least four months (*MKApplicable*) and are pairwise disjoint. C2 adds two measured properties, *monthsSinceLastCommit* (over the full history) and *locRelSenSlopeAnnual* (the annual Sen slope relative to the median LOC). *RecentlyActive* is the positive complement of *Dormant*, and *MinorLocGrowth* is the complement of *SubstantialLocGrowth*. Equation 2 shows one pair of regularity classes (the conjunct *SoftwareProject* is omitted).

The stage axioms of C1 and C2 are the following. Dormancy takes precedence, so PHASE_OUT needs no LOC data.

$$\begin{aligned}
\mathit{DeclineStage} &\equiv \mathit{MKApplicable} \sqcap \exists \mathit{locMKpValue.decimal}[< 0.05] \\
&\quad \sqcap \exists \mathit{locMKTau.decimal}[< 0] \\
\mathit{RBPhaseOutStage} &\equiv \mathit{Dormant} \\
\mathit{RBServicingStage} &\equiv \mathit{RecentlyActive} \sqcap \mathit{NonSubstantialChange} \\
\mathit{RBActiveDevelopmentStage} &\equiv \mathit{RecentlyActive} \sqcap \mathit{GrowthStage} \\
&\quad \sqcap \mathit{SubstantialLocGrowth}
\end{aligned} \tag{1}$$

Regularity classes are pairs of disjoint classes, for example for Law IV:

$$\begin{aligned}
\mathit{LawIVConformant} &\equiv \exists \mathit{monthsObserved.integer}[\geq 6] \\
&\quad \sqcap \exists \mathit{commitCountCV.decimal}[< 1.5] \\
\mathit{LawIVNonConformant} &\equiv \exists \mathit{monthsObserved.integer}[\geq 6] \\
&\quad \sqcap \exists \mathit{commitCountCV.decimal}[\geq 1.5] \\
&\quad \mathit{LawIVConformant} \sqcap \mathit{LawIVNonConformant} \sqsubseteq \bot
\end{aligned} \tag{2}$$

Unions such as *NonSubstantialChange* are written as one subclass axiom per operand. This entails the same memberships for individuals and avoids a reasoner crash that we observed with equivalent unions over datatype restrictions.

The regularity encodings reuse the thresholds of the earlier version. The identifiers *LawNConformant* and *LawNNonConformant* are internal names, read as "consistent with" and "not consistent with" the regularity. Law VI is a chained inference over the inferred C1 stage, and Law VIII requires that more than 30% of the months contain merges and more than 30% have several authors. Each law also carries a minimum-history guard.

Four anti-pattern classes combine the recent agent-commit share with LOC, complexity, dependency and refactoring growth. Only the copy-paste rule has all of its inputs in the reported populations. Complexity, dependency and refactoring history were not mined (revision decision D9, to keep mining affordable), so the other three rules cannot fire. No maintainer-published label exists for any anti-pattern, and the study design admits no human raters, so there is no answer key against which the rules could be checked. The agent-commit class has its inputs, but it serves only as an input to the

anti-pattern rules and is likewise not checked against an answer key; a dataset of known agent pull requests such as AIDev [5] could serve as one in future work. In the agent study, the "Inferred anti-patterns" heading of the history-based manifests read "none inferred" in 78 of the 80 manifests and listed copy-paste over reuse in the other 2. For the agent study, *StableModule* requires that less than 10% of a module's lines changed in the last 12 months and that it changed in at most 2 of those months, and the disjoint *ActiveModule* is the union of the two complements.

Table S1 lists Lehman's eight laws [6] and what SEO measures for each.

Table S1: Lehman's Laws and What SEO Checks

Law	**What SEO checks**
I Continuing change	Share of months with commits
II Increasing complexity	Mean complexity, first vs second half[a]
III Self-regulation	Coefficient of variation of monthly churn
IV Conservation of organisational stability	Coefficient of variation of monthly commits
V Conservation of familiarity	Coefficient of variation of monthly lines added[b]
VI Continuing growth	Inferred LOC-trend stage
VII Declining quality	Bug-fix ratio, first vs second half
VIII Feedback system	Months with merges and with several authors

[a] Undetermined for every repository, because complexity was not mined in the reported runs (Section 3.2). [b] Labelled but omitted from history-based manifests.

S1.4. Threshold Register

Every constant in the rules comes from a threshold register of 97 entries, each with an identifier, value, unit, rule, source, justification status and sensitivity values. Table S2 summarises it by group. Only one numeric value has a literature basis, the Mann-Kendall significance level $\alpha = 0.05$ [2, 3]. A second entry marked as justified is a method choice, not a numeric threshold (the original Mann-Kendall test, with the Hamed-Rao variant [7] as a sensitivity analysis). Every other entry is an unjustified convention, inherited from the earlier version or fixed a priori. The builder bakes the values into the axioms and annotates each class with the identifiers of its thresholds. A sensitivity analysis therefore rebuilds the TBox with an alternative value and

re-reasons the unchanged ABox, with no re-mining. Only the window length and the Mann-Kendall variant require re-measurement.

Table S2: Threshold Register Summary by Group

Group	**Entries**	**In TBox**	**Justified**	**Examples (value)**
C1 stages and Lehman regularities	17	16	1	α (0.05), minimum months (4), commit-count CV (1.5)
C2 Rajlich-Bennett stage rules	3	2	1[a]	D (12 months), G (0.05 per year)
Module stability	3	2	0	relative churn (0.10), change months (2)
Anti-patterns	15	15	0	recent agent share (0.25), LOC per feature (800)
Claim verification	6	6	0	consensus votes (2), status window (180 days)
Legacy ensemble stages (regression only)	22	22	0	τ vote (0.3), commit ratio (1.15)
Legacy law variant (regression only)	24	24	0	commit-count CV (0.45)
Earlier manifest generator (unused)	7	0	0	percentile (75)
Total	97	87	2	

[a] A method choice, not a numeric threshold: the original Mann-Kendall test, with the Hamed-Rao variant as sensitivity analysis.
Source: *ontology/modules/thresholds.yaml*. "In TBox" counts entries baked into an axiom. The entries outside the TBox are the mining window, the Mann-Kendall variant, the module-stability window (a measurement parameter) and the seven constants of the earlier generator.

S1.5. Claim Module Details

The claim type is asserted from the extraction, since it records what the manifest says and is input data, not a check result. The claim types are pairwise disjoint. The SENSO Miner asserts measured facts on each claim. For example, it counts the tracked paths equal to or ending with a claimed path, checks whether a *.gitignore* matches it, and counts the commits and changed lines inside a path over fixed windows. It also counts the declarations of a dependency across parsed manifests and computes the difference between a declared major version and the claimed one. A lookup is a measurement,

while deciding what its result means is inference. Every leaf class conjoins *CheckedClaim*, which requires a minimum number of extraction votes, so the consensus threshold is a TBox parameter. The ABox keeps every claim proposed by at least one of the three extraction models.

The claim strand of the running example (Section 3.3) is classified by the following axioms.

$$\begin{aligned} \mathit{DependencyMajorTooOld} \equiv\ & \mathit{VersionedDependencyClaim} \sqcap \mathit{DependencyDeclared} \\ & \sqcap \exists \mathit{declaredLowerMajorDelta}.\mathit{integer}[> 0] \\ \mathit{DependencyMajorTooOld} \sqsubseteq\ & \mathit{ContradictedClaim} \end{aligned}$$

Its justification contains three facts (the vote count, the exact-match count and the major-version difference) and eight axioms ending in $\mathit{DependencyMajorTooOld} \sqsubseteq \mathit{ContradictedClaim}$.

S1.6. Reasoners, Determinism and Storage

During development, Pellet was much slower than HermiT on the label modules, threw an internal exception on some of them and returned incomplete label sets that varied between runs (development logs in the replication package). Hence it is used only for stage A and for justifications. The stage A rules use SWRL string built-ins [8] to infer *AgentCommit* from co-author trailers, bot author names and branch prefixes, and *BugFixCommit* and *FeatureCommit* from the commit subject. Commits are reasoned in chunks of 1,000, which is sound because each label depends on one commit only. The inference and claims modules use datatype restrictions with facets, which none of the OWL 2 profiles allows [9], so OWL 2 DL is required.

The builder writes each module in a canonical order, so the TBox hash is identical across builds, and every inferred graph is named by that hash. Adding the module-stability classes for the agent study left the labels of all 120 RQ1 repositories unchanged. Asserted facts, labels and justifications are written to named graphs in an Apache Jena Fuseki triple store[1], with the reasoners, TBox hash and snapshot date as provenance. Every consumer reads labels from these graphs without recomputing them.

The differential test against the frozen Python oracle covered 338 scenario, boundary, random and grid-based cases. The four comparisons that did not

[1] `https://jena.apache.org/documentation/fuseki2/`

agree lie within 2×10^{-6} of a threshold, in legacy encodings and one anti-pattern rule. In 736 further comparisons, the old code returned a default label for insufficient data, while the ontology returned no label, because a missing measurement gives no label under the open-world assumption. Any comparison with numbers from the earlier version must account for these defaults.

S1.7. Reasoning Cost and Explanation Failures

Table S3 reports the reasoning cost. A repository-level HermiT call takes about 2 seconds, close to the 1.6 seconds that HermiT needs for an empty ABox, and a manifest-level call takes 4.5 seconds at the median. Stage A grows with the number of commits and is expensive only for very large histories. Explanations cost about as much as reasoning, because each call starts a new Java process.

Table S3: Reasoning Cost

Step	**Unit**	**Time (s)**
JVM start-up, HermiT, empty ABox	call	1.64
Differential test, 338 cases, 3 workers	total wall time	657.7
Stage A, Pellet, RQ1 repositories[a]	repository	median 2.66 (IQR 2.27–3.77), max 758.86
Stage B, HermiT, RQ1 repositories[a]	repository and configuration	median 2.07 (IQR 1.90–2.94)
Explanations, RQ1 repositories[a]	repository (median 9 calls)	median 22.89 (IQR 20.35–28.41)
Claims, HermiT	manifest	median 4.50 (IQR 3.63–6.77)
Claims, HermiT, 79 manifests	total wall time	170.2
Claim explanations, 99 targets	call	mean 9.35

[a] Computed from the 120 per-repository population records in the replication package; the maximum stage A time is a repository with 135,695 commits reasoned in 125 chunks.

In the RQ1 population, 9 of 1,042 explanation calls failed, all on one repository. In the agent study, explanation calls failed deterministically for some classes with decimal facets. When Pellet crashes on an explanation in the agent study, the call is repeated at most twice before the template fallback is used.

S1.8. Freeze Points and Model Panel

The RQ1 plan, including its thresholds and a decision table, was committed before any of the 120 sampled repositories was mined, and the git tag *rq1-freeze* then fixed the pipeline. For RQ2, the claim taxonomy, the thresholds and the extraction prompt were committed before any manifest was sent to a model. In the agent study, three git tags fixed the manifest generators, every run input, and the hashes of the outcome ledgers before any outcome was aggregated by condition.

Model panel. All model calls go to open-weight models on the NVIDIA NIM endpoint[2], and no paid model API is used. The RQ2 extraction panel has three models from three vendors and three architectures, chosen to reduce correlated errors. They are *nemotron-3-super-120b-a12b* (NVIDIA, mixture of experts), *gpt-oss-20b* (OpenAI open-weight, dense) and *diffusiongemma-26b-a4b-it* (Google, diffusion). A proposed claim survives only if its quote occurs verbatim in the manifest and models from at least two vendors propose it, so the models filter what is checked and never judge what holds. The agent study uses the first and the third as generator models. By October 2026, every model of our earlier study of the same manifest corpus [10] had been retired from the endpoint. This is one more reason why no result in the paper depends on a model's judgement.

S2. RQ1: Ground Truth, Sensitivity and Further Analyses

This section gives the detail behind the RQ1 results in Section 4.3.1 of the main paper. It covers the construction of the ground truth, the metrics and decision rules, the confusion matrices of all three classifiers, the ablation of the C2 rules, the sensitivity and noise analyses, and the convergent-validity and directional checks.

S2.1. Candidate Pool and Strata

The candidate pool combined the 10,000 most downloaded PyPI packages with 781 archived Python repositories found through the GitHub search API, giving 10,718 packages after merging. A candidate was kept if it had a GitHub repository that was not a fork, at least 24 months of history and a

[2] `https://integrate.api.nvidia.com/v1`, accessed through the OpenAI-compatible client.

PyPI release before the snapshot of 30 June 2025, which left 6,622 candidates. PyPI classifiers go stale, so the label is the classifier of the latest release published on or before the snapshot. Disagreeing signals are recorded as conflicts, and 941 candidates carried at least one. We drew 30 repositories per stratum with a fixed seed, from conflict-free candidates with at least 10 stars only (Table S4).

Close-down cannot be separated from phase-out with these signals, so it is not a stratum. Its 9 candidates in the sample stay in phase-out and are reported separately below. The labels carry known noise. In 63 of the 120 sampled repositories (52.5%), the classifier changed at least once in the release history, a quarter of the fetched releases carry no classifier, and 132 of 4,232 candidates (3.1%) had a classifier at the snapshot that differed from the one on their latest release.

Table S4: RQ1 Ground-Truth Strata at the Snapshot of 30 June 2025

Stratum	**Mapped signals**	**Eligible**	**Clean**	**Sampled**	**Class**
Initial dev.	Classifier 1 to 4	1,160	1,053	30	ACTIVE
Evolution	Classifier 5	2,151	2,060	30	ACTIVE
Servicing	Classifier 6; README maintenance mode	116	76	30	SERVICING
Phase-out	Classifier 7; archived; README unmaintained	993	399	30	PHASE_OUT

Eligible: candidates mapped to the stratum with at least 10 stars. Clean: eligible candidates with no conflicting signal. All sampled repositories are clean.

S2.2. Metrics and Decision Rules

The primary measure is Cohen's κ [11] over the 120 repositories. Weighted κ [12], macro-F1, balanced accuracy and per-class recall are secondary. Confidence intervals come from a stratified bootstrap with 10,000 draws, shared by all classifiers so that differences are paired. The pass, weak and fail decisions use 97.5% intervals, a Bonferroni split over C1 and C2, while other intervals are 95%. The pass cut-offs follow the bands of Landis and Koch [13], which are conventions rather than derived values [14]. The 12-month dormancy threshold and the 5% growth threshold of C2 are also conventions chosen in advance. A repository that gets no stage (*undetermined*) never counts as a match. In the decision table fixed in advance, a passing

C1 means GO, a passing C2 alone must be confirmed on a fresh holdout sample, and the decision is NO-GO if neither passes. CF is computed by separate Python code from monthly commit counts and does not share the trend inputs of C2.

S2.3. Agreement Metrics

Table S5 gives the exact values behind Figure 3 of the main paper.

Table S5: RQ1 Agreement With the Collapsed Ground Truth (n = 120)

Metric	**C1**	**C2**	**CF**
κ [95% CI]	0.04 [−0.05, 0.13]	0.49 [0.37, 0.61]	0.38 [0.28, 0.48]
κ 97.5% CI	[−0.07, 0.15]	[0.35, 0.62]	[0.26, 0.50]
Weighted κ	0.16 [0.06, 0.26]	0.52 [0.40, 0.64]	0.50 [0.39, 0.60]
Macro-F1	0.29 [0.23, 0.35]	0.67 [0.58, 0.75]	0.54 [0.47, 0.62]
Balanced accuracy	0.31 [0.25, 0.37]	0.70 [0.62, 0.78]	0.58 [0.52, 0.65]
Recall ACTIVE	0.75 [0.63, 0.85]	0.53 [0.40, 0.67]	0.75 [0.63, 0.85]
Recall SERVICING	0.13 [0.03, 0.27]	0.67 [0.50, 0.83]	0.10 [0.00, 0.23]
Recall PHASE_OUT	0.03 [0.00, 0.10]	0.90 [0.77, 1.00]	0.90 [0.77, 1.00]
Precision A / S / P	0.64 / 0.11 / 0.08	0.78 / 0.51 / 0.69	0.64 / 0.27 / 0.69
Undetermined	1	1	0
Pre-registered decision	FAIL	WEAK	not applicable

C1: Mann-Kendall stages; C2: lifecycle rule set; CF: commit-count comparator. Brackets: bootstrap intervals (10,000 stratified draws), 95% unless stated. The decision uses the 97.5% interval. Undetermined: no stage inferred, never counted as a match. Complete-case κ (undetermined dropped): C1 0.04, C2 0.49, CF 0.38.

S2.4. Measurement and Confusion Matrices

C1 maps the Mann-Kendall classes *GrowthStage*, *StabilizationStage* and *DeclineStage* to ACTIVE, SERVICING and PHASE_OUT. The Decline mapping is the claim the earlier model made, even though phase-out in Rajlich and Bennett's sense leaves the code unchanged rather than shrinking it. The 120 repositories were mined once at the *rq1-freeze* tag over a 36-month window up to the snapshot, and HermiT inferred the stage classes of C1 and C2 in one reasoning pass (stage B in Section 3.4 of the main paper). Before the labels were joined with the ground truth, they were compared with a Python oracle in all configurations, and the 2,880 comparisons showed no mismatch.

Figure S2 shows the confusion matrix of C2, and Table S6 gives the confusion matrices of all three classifiers. *DeclineStage* is inferred for only 13 repositories, one of them in phase-out, and 22 of the 30 phase-out repositories are classified as *StabilizationStage*. Servicing repositories are mostly classified as growing. For C2, 18 ACTIVE repositories are labelled SERVICING and 7 servicing repositories are labelled ACTIVE. The paired difference in κ between C2 and C1 is 0.45 (95% CI [0.33, 0.57]).

Table S6: RQ1 Confusion Matrices of C1, C2 and CF: Ground-Truth Strata by Predicted Label

	C1, unmapped phase				**C2**				**CF**		
Stratum	Gr	St	De	U	A	S	P	U	A	S	P
Initial development	19	7	3	1	14	10	5	1	22	3	5
Evolution	26	3	1	0	18	8	4	0	23	3	4
Servicing	18	4	8	0	7	20	3	0	24	3	3
Phase-out	7	22	1	0	2	1	27	0	1	2	27

Gr, St, De: *GrowthStage*, *StabilizationStage*, *DeclineStage* before the pre-registered mapping to A, S, P. A, S, P: ACTIVE, SERVICING, PHASE_OUT. U: undetermined. Each row sums to 30.

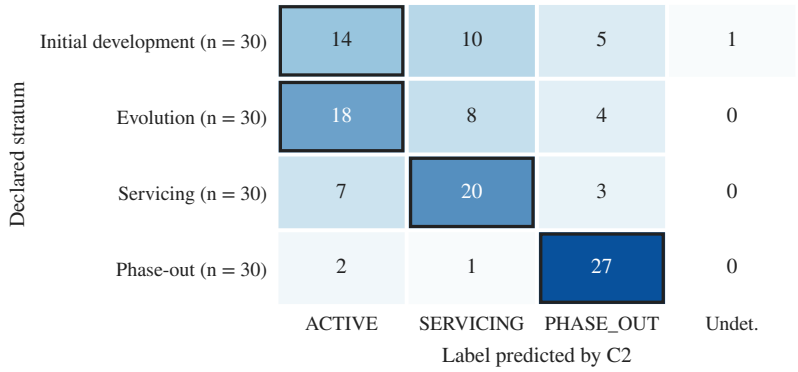


Figure S2: RQ1 confusion matrix of C2: declared ground-truth stratum (rows, 30 repositories each) by predicted label (columns; Undet.: no stage inferred). Cells give counts, darker cells hold more repositories, and outlined cells are correct predictions (initial development and evolution both map to ACTIVE). Phase-out is recovered well, and most errors fall between ACTIVE and SERVICING.

S2.5. Ablation of the C2 Rules

The pre-registered ablation ladder adds the C2 rules to C1 one at a time. Adding the dormancy rule alone raises κ to 0.46, a gain of 0.43 [0.33, 0.52], and lifts PHASE_OUT recall from 0.03 to 0.90. Adding the magnitude rule on top changes κ by only 0.02 [−0.07, 0.12]. However, it moves recall from ACTIVE (−0.18) to SERVICING (+0.33), so its main effect is to shift the boundary between those two classes. The difference between C2 and CF is 0.11 [−0.02, 0.23].

S2.6. Sensitivity and Ground-Truth Noise

Table S7 summarises the one-at-a-time sensitivity variants and the noise subsets, which by the pre-registered rule never change a pre-registered decision. C2 is weak in every variant, with κ from 0.34 (24-month window) to 0.49. C1 fails in every variant except one, in which *DeclineStage* is mapped to SERVICING and C1 becomes weak with $\kappa = 0.13$. The Hamed-Rao test [7] changes neither decision.

The variant with a 24-month dormancy threshold is not independent evidence, because the sampling treated "no push for 24 months" as a conflicting signal for a non-phase-out label. A descriptive grid of 288 configurations of C2 (computed by the Python oracle after it passed the differential test) gives κ between 0.24 and 0.49, and no configuration reaches the 0.60 point threshold.

The noise subsets drop repositories whose classifier changed in the year before the snapshot (N1), whose classifier at the snapshot differs from the latest one (N2), or whose label came from undated README text (N3). If label noise were hiding a valid model, agreement should rise as noisy labels are removed. It does not, since C2 stays between 0.45 and 0.53 and remains weak in every subset, and C1 fails in every subset. In the descriptive close-down split, C2 recovers all 9 close-down candidates as PHASE_OUT and 18 of the other 21 phase-out repositories. C1 labels all 9 close-down candidates as SERVICING.

S2.7. Convergent Validity and Directional Contrasts

Convergent validity compares the classifiers with each other. C1 and CF agree only weakly ($\kappa = 0.11$, 95% CI [0.01, 0.20]). C2 and CF agree moderately over all repositories ($\kappa = 0.46$, [0.36, 0.57]). However, on the 81 repositories that neither labels PHASE_OUT, their agreement drops to −0.07 [−0.21, 0.07]. The agreement between C2 and CF therefore comes entirely

Table S7: RQ1 Sensitivity Variants and Ground-Truth Noise Subsets

ID	**Change from primary**	**n**	**C1 κ [97.5% CI]**	**C2 κ [97.5% CI]**
Primary	none	120	0.04 [−0.07, 0.15] F	0.49 [0.35, 0.62] W
S1	$\alpha = 0.01$	120	0.08 [−0.03, 0.20] F	0.49 [0.35, 0.62] W
S1	$\alpha = 0.10$	120	0.02 [−0.08, 0.13] F	0.49 [0.35, 0.62] W
S2	window 24 months	120	−0.01 [−0.12, 0.10] F	0.34 [0.21, 0.46] W
S2	window 48 months	120	0.08 [−0.02, 0.19] F	0.47 [0.33, 0.61] W
S3	D = 6	120	n/a	0.44 [0.31, 0.57] W
S3	D = 18	120	n/a	0.44 [0.29, 0.58] W
S3	D = 24†	120	n/a	0.40 [0.25, 0.54] W
S4	G = 0	120	n/a	0.46 [0.33, 0.60] W
S4	G = 0.02	120	n/a	0.45 [0.31, 0.59] W
S4	G = 0.10	120	n/a	0.38 [0.27, 0.49] W
S5	Hamed-Rao test	120	0.05 [−0.07, 0.17] F	0.47 [0.33, 0.60] W
S6	Decline to SERVICING	120	0.13 [0.01, 0.26] W	n/a
N1-lenient	classifier changed in year	117	0.06 [−0.05, 0.17] F	0.50 [0.37, 0.63] W
N1-strict	also unverifiable history	96	0.03 [−0.09, 0.15] F	0.53 [0.39, 0.66] W
N2	stale classifier	111	0.05 [−0.06, 0.17] F	0.50 [0.37, 0.63] W
N3	README-sourced label	111	0.03 [−0.07, 0.14] F	0.45 [0.32, 0.59] W
N-clean	N1-strict, N2 and N3	82	0.04 [−0.08, 0.16] F	0.48 [0.34, 0.62] W

Primary: $\alpha = 0.05$, 36-month window, original Mann-Kendall test, D = 12, G = 0.05. D: dormancy threshold in months; G: minimum relative LOC growth per year. F: FAIL; W: WEAK. Noise subsets drop the repositories named in the Change column. †Not independent evidence (see text). CF κ ranges from 0.26 to 0.41 across all rows.

from the recency rule they share, and the LOC trend and the commit trend carry no common signal about the ACTIVE and SERVICING boundary.

Table S8 reports the pre-registered directional contrasts between the collapsed classes, with Holm correction [15] over E1 to E6 and Cliff's δ [16] as effect size. Two contrasts survive correction. Servicing repositories are active in more months than phase-out repositories (E2), and ACTIVE repositories grow faster relative to their size than servicing ones (E3), which supports the direction of C2's magnitude rule. The remaining contrasts, including the expected higher bug-fix ratio in servicing (E4), are not supported. The median number of months since the last commit is 43.5 for phase-out repositories, against 0 to 2 for the other strata.

The exploratory contrasts X1 and X2 support the merge of initial development and evolution, since neither growth rate nor activity differs measurably between the two strata, and C2 labels 47% of the initial-development and

Table S8: RQ1 Directional Contrasts Between Ground-Truth Classes

ID	Contrast	n_x	n_y	Cliff's δ [95% CI]	Holm p
E1	*activeMonthRatio* A > S	60	30	0.18 [−0.09, 0.43]	0.26
E2	*activeMonthRatio* S > P	30	30	0.75 [0.56, 0.91]	1.3×10^{-6}
E3	*locRelSenSlopeAnnual* A > S	59	30	0.31 [0.07, 0.54]	0.04
E4	bug-fix ratio S > A	28	56	0.25 [−0.02, 0.50]	0.13
E5	feature-commit share A > S	19	7	0.08 [−0.52, 0.65]	0.78
E6	commit Mann-Kendall τ A > S	60	30	−0.01 [−0.27, 0.25]	0.78
X1	*locRelSenSlopeAnnual* ID vs EVO	29	30	−0.17 [−0.46, 0.12]	0.25*
X2	*activeMonthRatio* ID vs EVO	30	30	−0.23 [−0.51, 0.06]	0.12*

One-sided Mann-Whitney tests for E1 to E6; A, S, P are the collapsed ground-truth classes. *X1 and X2 are exploratory, two-sided and uncorrected. Smaller n means the measure was undefined for some repositories.

60% of the evolution repositories ACTIVE (difference −0.13 [−0.37, 0.10]).

S2.8. Further Results Moved From the Main Text

C1 labels 22 of the 30 phase-out repositories as stabilising, as the pre-registration predicted, because frozen code has a flat LOC series without a trend, and it labels servicing repositories mostly as growing, because a steady stream of small patches gives a nearly monotone LOC series. For C2, most errors fall between ACTIVE and SERVICING (Figure S2), and a further 9 ACTIVE repositories are labelled PHASE_OUT because they had no commit for a year while their classifier still declared an active status. Adding the dormancy rule alone to C1 raises κ to 0.46 and PHASE_OUT recall from 0.03 to 0.90, while the growth rule adds only 0.02 [−0.07, 0.12]. On the 81 repositories that neither C2 nor CF labels PHASE_OUT, the two agree at chance level ($\kappa = -0.07$). C2 stays weak in every sensitivity variant (κ between 0.34 and 0.53), and C1 fails in every variant except one that maps decline to SERVICING; removing the noisiest labels does not raise agreement. Per-class claims are allowed only where recall has a 95% lower bound of at least 0.50, which for C2 holds for SERVICING and PHASE_OUT, not for ACTIVE. In an exploratory re-analysis weighted by population, C2 drops to $\kappa = 0.25$ against 0.37 for CF. RQ1 does not test whether Lehman's laws hold.

S3. RQ2: Claim Extraction, Sensitivity and First-Pass Detail

This section gives the detail behind the RQ2 results in Section 4.3.2 of the main paper. It covers snapshot resolution, the flow of manifests and claims through extraction, the agreement between the three extraction models, the sensitivity variants and the leaf-level comparison of the first and final passes.

S3.1. Corpus and Snapshot Resolution

The corpus reuses the 80 repositories and manifest texts of our earlier conference study, which classified manifest sections into topic categories with a vote of three language models. Everything else is new, namely statement-level claims, the commit at which each manifest was read, repository measurements, reasoner check results with justifications, a separately implemented re-check, a second-instrument triangulation, a drift analysis and the extraction panel. Of the 79 manifests that entered the analysis, 37 are *CLAUDE.md* files, 29 *AGENTS.md*, 12 *copilot-instructions.md* and 1 a *.clinerules* file. Six manifests written mostly in a CJK script were kept, because the extraction checks do not depend on language.

The cached manifest texts were fetched without recording a commit. The snapshot S of a repository is therefore the newest first-parent commit before the date on which the corpus was committed (March 2026) whose manifest text equals the cached text. The manifest and the repository are checked at the same commit S, and every history window ends at S. The text matched at the upper bound for 77 manifests and at an earlier version for 2, and 1 manifest had no matching version and was excluded.

S3.2. Claim Extraction and Flow

Each manifest was split at its headings into chunks of at most 6,000 characters, which gave 152 chunks, and each chunk was sent to the three panel models with temperature 0. The prompt defines the claim types with synthetic examples only, so no corpus text leaks into the prompt. It asks for the verbatim quote, the subject, the line and a mode. Only asserting claims are checked.

Every proposal then passes mechanical checks. The response must parse against a schema, the quote must occur verbatim in the chunk after whitespace and markup normalisation, and the subject must occur in the quote. Type-specific shape filters reject placeholders, URLs, non-identifier package names and similar artefacts. Each surviving claim is reduced to a canonical

key, such as the normalised path for T1, so that proposals from different models match exactly. A claim enters the ABox with its number of votes, and *CheckedClaim* requires at least two votes, so the consensus rule is a TBox threshold that can be varied without re-extraction.

Table S9 traces manifests and claims through these steps. For 500 *diffusiongemma-26b-a4b-it* claims the model omitted the quote, which was repaired mechanically from the subject's line, and a strict variant that rejects these claims is reported alongside. After consensus and the exclusion of T7 claims and of status claims that only protect generated files, 1,219 asserting claims remained in the 79 analysed manifests, and every one of them received a measurement and a check result.

Table S9: RQ2 Flow of Manifests and Claims

Manifests			
Corpus / cloneable / snapshot resolved			80 / 80 / 79
text equal at upper bound / earlier version / no match			77 / 2 / 1
With at least one checked claim			68
Inconsistent ABoxes			0
Extraction per model	**N**	**O**	**G**
Responses (parse failures)	152 (10)	152 (2)	152 (3)
Proposed claims	2,866	1,909	2,455
Rejected by grounding / by shape	116 / 1,027	304 / 434	321 / 810
Accepted (of which quote repaired)	1,837 (0)	1,394 (0)	1,457 (500)
Consensus			
Canonical units, union of models (after 91 path merges)			1,994 (1,903)
Units with 1 / 2 / 3 votes, after merges			653 / 685 / 565
Units with at least 2 votes / of which asserting			1,250 / 1,232
Asserting, at least 2 votes, in analysed manifests			1,228
excluded: T7 or protected-status claims			9
Checked claims, measured, with a check result			1,219

N: *nemotron-3-super-120b-a12b*; O: *gpt-oss-20b*; G: *diffusiongemma-26b-a4b-it*.

S3.3. Extraction Agreement

Table S10 reports the agreement between the three models. The mean pairwise F1 over canonical keys is 0.61 (0.53 in the strict variant), highest for

commands and lowest for path status. Krippendorff's α [17, 18] needs a fixed set of units, which extraction does not have, so two bounding versions were computed. Presence over the union of keys gives a lower bound of $\alpha = -0.03$ (95% CI [−0.08, 0.02]), deflated by the many units that only one model proposed, and treating every manifest line as a unit gives an upper bound of $\alpha = 0.53$ [0.45, 0.60]. Agreement on the mode attribute is at chance level ($\alpha = -0.01$).

Table S10: RQ2 Extraction Agreement Between the Three Panel Models

	Pairwise F1			**Presence**
Claim type	**N, O**	**N, G**	**O, G**	$\boldsymbol{\alpha}$
T1 path existence	0.59	0.67	0.51	−0.04
T2 path status	0.46	0.38	0.36	−0.14
T3 tool use	0.62	0.75	0.63	−0.04
T4 dependency	0.59	0.46	0.40	−0.18
T5 command	0.80	0.84	0.76	−0.03
T6 version	0.63	0.42	0.43	−0.12
All types	0.62	0.67	0.55	−0.03
All types, strict (no quote repair)	0.62	0.53	0.45	−0.10

N, O, G: as in Table S9. The N, O pair is unaffected by quote repair. Presence α over the union of keys is a lower bound. Line-level α (upper bound): 0.53 [0.45, 0.60], strict 0.46 [0.37, 0.53]. Mode α: −0.01 (549 units). Regex recall of the consensus (of the union): T1 0.56 (0.77) of 665 candidates; T5 0.84 (0.94) of 219.

Two mechanical checks bound the remaining extraction error. A regular-expression extractor finds candidate paths and commands in code spans independently of the models. The consensus captures 56% of the path candidates and 84% of the command candidates, so a sizeable share of the paths written in code spans is never checked. A cue-word check bounds mode errors, since only 2 of the 36 contradicted T1 claims contain a word such as "example" or "create" in their quote. The low mode agreement is also addressed by a sensitivity variant (Section S3.6).

S3.4. Check Result Details

Table S11 gives the exact counts and rates behind Figure 4 of the main paper, and its note lists the counts per kind of contradiction shown in Figure S3.

At the manifest level, the 29 manifests with a contradiction are those inferred to be members of *ManifestWithContradiction*, the chained class for

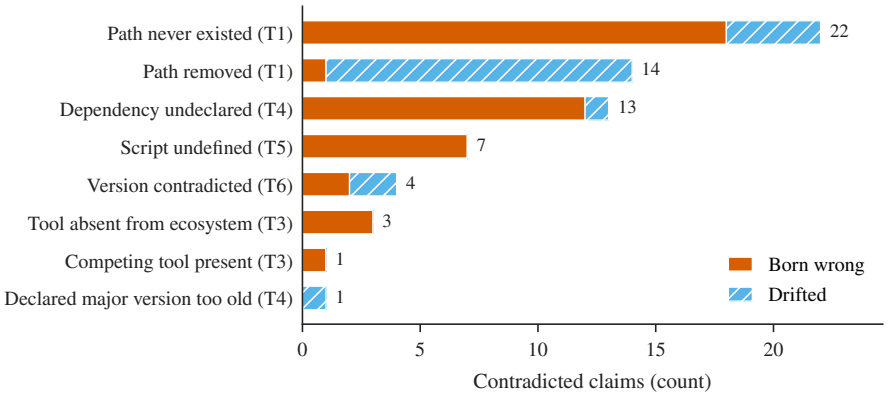


Figure S3: Kinds of contradicted claims (n = 65), split into claims that were already false when added to the manifest (born wrong) and claims that became false later (drifted).

a manifest that makes at least one contradicted claim. The per-manifest distribution is skewed, because the median manifest has no contradicted claim, and the upper quartile has a rate of 7.7%. An exploratory correlation between the days since a manifest was last changed and its contradiction rate is weak and not significant ($\rho = 0.14$, $p = 0.24$, n = 68). For drifted claims, the commit that made the claim false was located by bisection, assuming a single change of truth value.

S3.5. Validation of Check Results

Table S12 gives the re-check, the second-instrument triangulation and the drift analysis behind Section 4.3.2 and Section 4.3.2 of the main paper. Against the re-check, the reasoner's contradicted result has a precision of 0.98 and a recall of 1.00. For tools and dependencies, a whole-tree *git grep* counts as corroboration only when it finds nothing. Where the name occurs, this may point to a gap in the dependency parsers, an alias or an incidental mention.

S3.6. Threshold and Consensus Sensitivity

Table S13 varies one design parameter at a time by rebuilding the TBox and re-reasoning over unchanged ABoxes. The T2 thresholds barely matter, because only four T2 claims received a check result. Accepting claims proposed by a single model raises the number of contradicted claims from 65 to 162 and of affected manifests from 29 to 40, and requiring all three models

Table S11: RQ2 Check Results by Claim Type and Contradiction Rates

Type	n	Sup.	Con.	Unv.	I/K	Pooled rate % [95% CI]	Manifests with ≥1 % [95% CI] (n)
T1 path	839	772	36	31	0/0	4.5 [2.4, 7.0]	27.7 [16.9, 38.5] (65)
T2 status	4	3	0	0	1/0	0	0 (2)
T3 tool	100	96	4	0	0/0	4.0 [0.9, 8.5]	8.3 [2.1, 16.7] (48)
T4 dependency	65	51	14	0	0/0	21.5 [13.6, 30.0]	58.8 [35.3, 82.4] (17)
T5 command	189	181	7	1	0/0	3.7 [0.0, 9.7]	6.7 [0.0, 16.7] (30)
T6 version	22	12	4	3	0/3	21.1 [5.6, 36.8]	23.5 [5.9, 47.1] (17)
All types	1,219	1,115	65	35	1/3	5.5 [3.6, 7.7]	42.6 [30.9, 54.4] (68)

Sup., Con., Unv., I/K: Supported, Contradicted, Unverifiable, Indeterminate / Conflicted. Rate: Contradicted / (Supported + Contradicted + Conflicted). n in the last column: manifests with at least one decidable claim of the type. Kinds of contradiction: T1 path never existed 22, path removed 14; T3 tool absent from the ecosystem 3, competing tool present instead 1; T4 dependency undeclared 13, declared major version too old 1; T5 script undefined 7; T6 version contradicted 4. Supported T1 claims include 170 paths found only by suffix. Manifest-level CIs from 10,000 bootstrap resamples of manifests. Mean per-manifest rate over all types: 6.7% [4.1, 9.7]. The T3 and T4 rates are upper bounds (Section 4.3.2 of the main paper).

lowers them to 23 and 14. However, the mean per-manifest rate stays between 6.7% and 8.5%, so the consensus threshold changes how many claims are checked much more than the rate per manifest. Excluding the *PathNeverExisted* subtype gives a manifest share of 29.4% [19.1, 39.7], and keeping only mode-unanimous claims gives 23.3% [13.3, 35.0] of 60 manifests.

S3.7. First-Pass and Final Leaf Counts

The fixes after the first pass stopped treating import statements, aliases and dotted package names as paths, kept glob characters and the root folder of tree diagrams, required the runner and target of a command to occur together in the quote, tightened the shape of dependency subjects, added parsers for further dependency-manifest formats and added one missing test-framework signature. No check rule or threshold changed. Table S14 compares the two passes leaf by leaf. Most of the reduction comes from *PathNeverExisted* and *DependencyUndeclared*, the kinds most exposed to extraction errors. Some fixes also added contradictions, as *PathRemoved* rose from 10 to 14. The first-pass total of 113 contradicted claims is a sum over a different claim set, so the two passes are not a paired comparison (Section 4.3.2 of the main paper).

Table S12: RQ2 Validation of Check Results: Re-check, Triangulation and Drift

	Re-check		Triangulation		Drift
Type	**n**	**Agree (κ)**	**Instrument**	**Corroborated**	**Drifted / born wrong**
T1	839	1.00 (1.00)	Contents API	36/36 C, 50/50 S	17 / 19 (13 d)
T2	12	1.00 (1.00)	Commits API	1/1 I	n/a
T3	100	1.00 (1.00)	*git grep*	1/4 C	0 / 4
T4	65	0.98 (0.95)	*git grep*	2/14 C	2 / 12 (113 d)
T5	189	1.00 (1.00)	CI files	7/7 C	0 / 7
T6	22	1.00 (1.00)	Second parser	22/22	2 / 2 (17.5 d)
All	1,227	0.999 (0.995)			21 / 44 (13 d)

C, S, I: Contradicted, Supported, Indeterminate claims checked. Re-check n includes 8 protected-status claims not in the headline rates. Re-check precision of Contradicted 0.98 (64 of 65), recall 1.00. Drift counts are over contradicted claims; in parentheses the median number of days between the commit that made a drifted claim false and the snapshot.

S3.8. Fresh Check of 30 New Repositories

This subsection gives the detail behind Section 4.3.2. The plan (*papers/paper1/wp2c-fresh-check-design.md* in the replication package) was committed before any run. GitHub code search for root *CLAUDE.md* and *AGENTS.md* files returned 1,868 candidates, of which 120 were public, not forks, not archived, had at least 20 stars, had been pushed since 1 July 2026, had a manifest of at least 500 characters and were neither RQ2 nor tool-development repositories. They were shuffled with a fixed seed and checked in order until 30 runs completed; no run failed. The tool used the RQ2 rules, prompts, models and TBox, with one fix made before the runs: a path claim whose quote shows a wildcard (`*`, `*/` or `**/`) directly before the subject is treated as a pattern. Applied offline to the 1,371 committed RQ2 path claims, the fix changes none. Runs took a median of 88 seconds (mean 142, maximum 1,199) with two at a time.

Of the 553 claims with a result, 486 were supported, 41 contradicted, 25 unverifiable and 1 conflicted; 38 contradicted claims are paths (33 *PathNeverExisted*, 5 *PathRemoved*) and 3 are versions. The planned instruments (GitHub Contents API for paths, an independent re-parse for versions) and a check that the subject is not a wildcard, a URL or followed by a placeholder passed 40 of the 41 (97.6%, Wilson 95% interval 87.4 to 99.6). Because these checks test only that the named path or version is absent, the authors also read every contradicted claim next to the repository at its snapshot, as in RQ2's first pass. This step was added after the runs and is

Table S13: RQ2 Sensitivity of Contradiction Counts to Design Choices

Variant	**Con.**	**Man.**	**Rate**
Baseline (at least 2 votes)	65	29	pooled 5.5% [3.6, 7.7]; mean 6.7%
Consensus at 1 vote	162	40	mean 8.5% (71 evaluable)
Consensus at 3 votes	23	14	mean 7.0% (60 evaluable)
Mode agreed by all three models (545 claims)	23		pooled 4.3% [2.0, 7.2]
Foreign-manifest subtype excluded	43		pooled 3.7% [2.1, 5.4]
T2 window 90 days; T2 contradiction thresholds	65		unchanged
T2 window 365 days	66		one T2 claim contradicted
T2 support threshold 0 commits	65		one more Indeterminate
First pass, before normalisation fixes	113	38	see Table S14

Con.: contradicted claims. Man.: manifests with at least one contradicted claim, out of 79; blank where not reported for the variant. Mean: mean per-manifest rate. Foreign-manifest subtype: *PathNeverExisted*.

descriptive. Table S15 gives the result; each label and its evidence is in *scripts/wp2c_inspection.py*. No tool change was made after the runs. The largest kinds of false alarm point to fixes for the extraction step: split slash alternations, read Go module paths as packages, and skip absolute and template paths.

S3.9. RQ3: Persistence of Contradicted Claims

This subsection gives the detail behind Section 4.3.3. The design was pre-registered as part of RQ2 (there called RQ2b) and is numbered RQ3 in the paper. The latest commit of the default branch was fetched once for the 68 repositories whose manifests hold a supported or contradicted claim of types T1 and T3 to T6. No repository was unavailable, 3 were archived, and the snapshot was an ancestor of the latest commit in every repository. Fourteen repositories had no commit after the snapshot, 36 manifests had been edited, and 3 manifests had been deleted. A claim is kept when one line of the manifest at the latest commit still contains every component of its key, such as the path or the script name. For 76 path claims drawn from tree diagrams, the key is the last component of the path. Re-running the measurement and the re-check at the snapshot reproduced all 1,177 frozen results, and the re-check agreed with HermiT on all 1,061 kept claims at the latest commit.

Table S14: RQ2 Leaf Counts Before and After the Normalisation Fixes

Check result	**Leaf class**	**First pass**	**Final**
Contradicted	*PathNeverExisted*	57	22
	PathRemoved	10	14
	DependencyUndeclared	25	13
	DependencyMajorTooOld	1	1
	ScriptUndefined	10	7
	ScriptHostMissing	1	0
	ToolAbsentInEcosystem	4	3
	ToolAbsentCompetitorPresent	1	1
	VersionContradicted	4	4
	Total contradicted claims	113	65
Supported	*PathFoundExact*	553	602
	PathFoundBySuffix	165	170
	DependencyConfirmed	67	36
	DependencyMajorInRange	23	15
	ScriptDefinedRoot	186	171
Manifests with contradiction		38	29

Table S17 lists the analyses. The pre-registered wording for A1 is "most persist". In the edited manifests, the cluster bootstrap interval is wide, and the pre-registered wording there is "about half persist; the data cannot say whether most do". The pooled ratio is close to 1 because manifests without contradicted claims had more removals of supported claims, which the within-manifest comparison holds constant. Of the 74 removals, 70 replaced the line with other text and 4 deleted it, and 8 were made in commits with an agent signature. Seven deviations were logged: the analysed set has 68 manifests rather than the 79 stated in the design, two fixes to the key matcher and the rule for tree-derived paths were made before the latest commits were fetched, and the remaining entries record implementation details and a smaller bootstrap for the conditional logistic regression.

S4. RQ4 and RQ5: Agent Study Details

This section gives the details of the agent study that the main paper summarises in Section 4.3.4. It covers task sampling, run settings, the secondary outcomes, the estimators and the power analysis, stage-specific estimates, condition B and uptake, and the sensitivity analyses.

Table S15: Contradicted Claims in the Fresh Check of 30 New Manifests, by Inspection

Kind	**Claims**	**Manifests**
Real errors		
File or folder never in the repository	7	4
File or folder removed from the repository	5	2
Version contradicted by *go.mod*	2	1
Total real errors	14	5
False alarms (extraction)		
Two files written as one (*src/elf.c/h*)	17	1
Go import or package path read as a file path	5	2
Naming template, path outside the repository, build output, repository name as tree root, version of another tool	5	5
Total false alarms	27	7

Table S16: RQ3 Outcomes at the Latest Commit by Check Result at the Snapshot

At the snapshot	**Manifest deleted**	**Removed**	**Still contradicted**	**Now supported**	**Total**
Contradicted	0	10	52	3	65
born wrong	0	6	36	2	44
drifted	0	4	16	1	21
Supported	42	64	9	997	1,112

S4.1. Task Sampling and Strata

The tasks come from *SWE-rebench* [19], a continuously collected set of SWE-bench-style tasks [20]. Each task fixes a base commit, a problem statement, a gold patch, a test patch, and lists of fail-to-pass (F2P) and pass-to-pass (P2P) tests. *SWE-rebench* was chosen because it covers enough repositories to supply tasks in servicing and phase-out. The stage of each task is the maintainer-declared stage at the base date, read from the PyPI *Development Status* classifier in effect at the base commit (2 to 4 initial development, 5 evolution, 6 servicing, 7 phase-out). The declared stage was fixed as the stage variable whatever RQ1 concluded. It is independent of the treatment, because condition C shows the inferred stage and never the declared one.

Starting from 21,336 tasks, we removed tasks that could not be run, tasks whose repository is not on PyPI, and tasks without a classifier at the base date. With a cap of 8 tasks per repository, the stratified pool held 3,292 tasks from 2,347 repositories, and phase-out and servicing tasks are rare in

Table S17: RQ3 Primary and Robustness Analyses (A1: Share Still Contradicted; A2: Risk Ratio of Removal, Contradicted Versus Supported)

Analysis	**A1 [95% CI]**	**A2 risk ratio [95% CI]**
Primary (Wilson CI; Mantel-Haenszel within manifest)	0.80 [0.69, 0.88]	3.59 [1.59, 11.32]
Cluster bootstrap over manifests	0.80 [0.65, 0.93]	
Manifests edited after the snapshot	0.71 [0.48, 0.91]	3.59 [1.62, 10.52]
Without archived repositories	0.79	3.59 [1.58, 10.92]
Reasoner and re-check agree	0.80	3.59 [1.59, 11.32]
Removal means the quoted text disappeared		2.46 [1.20, 3.93]
Pooled over all manifests		1.61 [0.31, 4.03]
Conditional logistic regression (odds ratio)		9.14 [2.96, 28.69]
GEE, exchangeable (odds ratio)		2.36 [1.13, 4.94]

it (Table S18). A task entered the sample only after validation, in which the environment was built, the F2P tests failed at base and passed with the gold patch, and the P2P tests passed in both states. Of 148 attempted tasks, 113 passed validation, and 107 remained after the 6 tasks used during development were excluded. The pre-registered rule caps the sample at 80 and keeps every phase-out task, which gives 80 tasks from 61 repositories, including all 31 phase-out tasks.

Table S18: Agent-Study Task Strata, Supply and Condition B Availability

Stratum	**PyPI status**	**Pool**	**Validated**	**Sampled**	**Repos**	**B avail.**	**H3 level**
Initial development	2 to 4	1,434	26	16	16	7	ACTIVE
Evolution	5	1,732	25	16	16	9	ACTIVE
Servicing	6	91	25	17	16	6	SERVICING
Phase-out	7	35	31	31	13	15	PHASE_OUT
Total		3,292	107	80	61	37	

For H3, stages are collapsed as in RQ1. The plan admits phase-out as a separate H3 level only if it has at least 10 tasks from at least 4 repositories. This holds with 31 tasks from 13 repositories, so H3 has two numerator degrees of freedom. Condition B exists for 37 of the 80 tasks.

S4.2. Agent, Models and Run Settings

The agent is *mini-SWE-agent*[3] in its text mode, a minimal loop from the authors of *SWE-agent* [21] in which the model issues one shell command

[3] `https://github.com/SWE-agent/mini-swe-agent`

per step and reads its output. Each run starts in a fresh workspace that contains only the ancestry of the base commit, without the F2P tests. In-tree contributor or agent files are left in place in every condition. A run ends when the agent submits, reaches the step limit of 40, exceeds the wall-time limit of 1,800 seconds, or produces repeated format errors.

The two models, *nvidia/nemotron-3-super-120b-a12b* and *google/diffusiongemma-26b-a4b-it*, ran with temperature 0.7 and top-p 0.95. They were the only models that completed the agent loop in a feasibility study on a toy repository, so they are not a sample from a population of models. Run settings were adjusted after a pilot from failure modes and the condition A baseline only, and then frozen. Every task is run under conditions A, C and D, and under B when available, with 2 models and 2 replicates per condition, in a seeded random order.

The appended section has the fixed heading *Repository context* and at most 8,000 characters. Condition B concatenates the repository's own contributor and agent files at the base commit (for example *CONTRIBUTING*, *CLAUDE.md*, *AGENTS.md* or *.github/copilot-instructions.md*) with path headers, and excludes README files. Condition C shows the inferred lifecycle stage and the LOC-trend class with their filled explanation templates, one line per evolution regularity (Laws I to IV and VI to VIII, with Law II always undetermined) stating whether the repository is consistent with it, the inferred anti-patterns, a stability label for each of the largest modules, and fixed notes. Module stability is a relative-churn measure in the sense of Nagappan and Ball [22]. Because RQ1 ended in NO-GO, C adds a fixed sentence saying that the stage label has not been shown to agree with maintainer-declared status, and C contains no imperatives. Condition D lists the same measured properties as rows of name, value and definition, plus raw module measurements and monthly activity counts, with no labels, thresholds, comparisons, explanations or rankings.

Before the run freeze, mechanical checks confirmed that generation is deterministic, that D contains no interpretive words and C no imperatives, that every number in C also appears in D, and that D is equal in length to C within 10%. Neither contains the declared PyPI status or anything from the gold or test patch. The manifests were injected into the prompt and not written to a file, because in the feasibility study an agent given a *CLAUDE.md* file in the repository root saw the file and never opened it.

S4.3. Resolve Rates and Confirmatory Tests

Tables S19 and S20 give the exact values behind Figure 6 of the main paper and Figure S4 of the main paper. All 1,108 scheduled runs reached a valid end state, and no run failed for infrastructure reasons. Pooled over models, resolve rates lay between 0.12 and 0.17 in every condition, and the order of C and A is reversed between the two models. The C versus A risk difference from the task-level bootstrap is less than one percentage point.

Table S19: Resolve Rates and Terminal States by Condition and Model

		Resolved / runs (rate)			Terminal state			
Cond.	**Runs**	**Both models**	**nemotron**	**diffusiongemma**	**Subm.**	**Step lim.**	**Format**	**Wall**
A	320	53 (0.166)	32/160 (0.200)	21/160 (0.131)	154	163	3	0
C	320	55 (0.172)	39/160 (0.244)	16/160 (0.100)	127	188	5	0
D	320	43 (0.134)	28/160 (0.175)	15/160 (0.094)	117	200	2	1
B[a]	148	18 (0.122)	12/74 (0.162)	6/74 (0.081)	56	92	0	0

[a] B is run only on the 37 tasks where it exists and is not part of the confirmatory analysis.

Table S20: Confirmatory Hypotheses H1 to H3 (GEE, 80 Tasks, 61 Repositories, 960 Runs)

Hyp.	**Contrast**	**Estimate [95% CI]**	**Raw p**	**Holm p**	**Risk diff. [95% CI]**	**Test result**
H1	C vs A	OR 1.05 [0.77, 1.42]	0.7713	1	0.0063 [−0.0375, 0.0437]	not rejected
H2	C vs D	OR 1.34 [1.02, 1.74]	0.0332	0.0997	0.0375 [0.0031, 0.0719]	not rejected
H3	C vs A × stage	$F(2, 60) = 0.03$	0.9669	1	n/a	not rejected

OR: marginal odds ratio. Risk differences from the task-level paired bootstrap (10,000 resamples, stratified by declared stage).

S4.4. Secondary Outcomes

Every outcome is computed from the agent's final diff without any LLM. The pre-specified secondary outcomes are defined on the non-test part of the diff, as follows:

- Y0 - a non-empty non-test patch was produced.
- Y1 - the change in cyclomatic complexity summed over touched functions, measured with *lizard*[4].

[4] `https://github.com/terryyin/lizard`

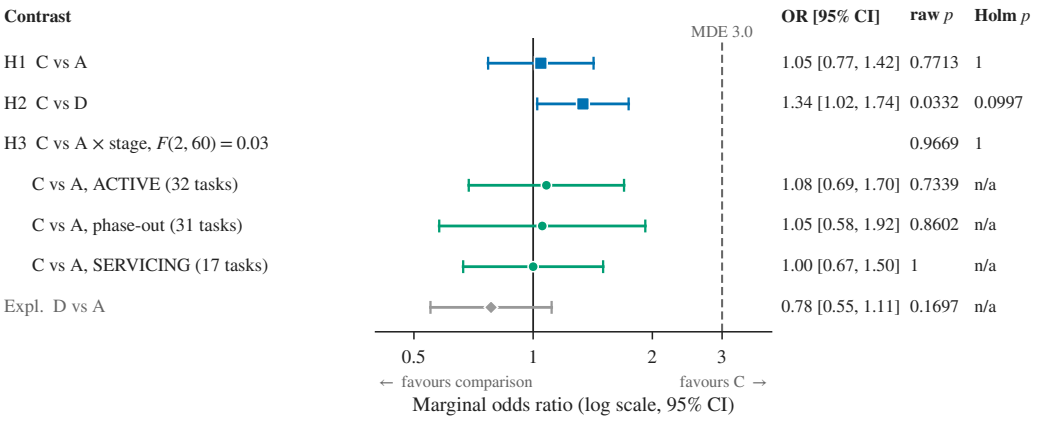


Figure S4: Confirmatory hypotheses H1 to H3 (GEE, 80 tasks, 61 repositories, 960 runs). Marginal odds ratios with 95% confidence intervals on a log scale. The solid line marks no effect and the dashed line the minimum detectable effect (odds ratio 3.0). H3 is an F-test with no single odds ratio, so the stage-specific estimates beneath it and the exploratory D versus A contrast are descriptive. Holm adjustment applies to H1 to H3 only.

- Y2 - the number of new import edges between modules of the partition used for module stability.
- Y3 - the share of changed lines that fall in modules labelled stable, with a threshold-free variant (Y3w) and the excess over the gold patch (Y3_excess).
- Y4 - the diff size relative to the gold patch, as $\log_2$ of the ratio of changed lines.
- Y5 - the number of files touched.
- Y6 - the number of steps used, with reaching the step limit as a binary companion.

Y3 is partly a manipulation check, because it uses the labels that C displays, and Y1, Y2 and Y4 are defined only when Y0 holds. As a descriptive check of uptake, we recorded whether the agent's messages mention *repository context*, a C label term, or a D display name. Table S21 gives the C versus A and C versus D contrasts. Only the differences in Y4 and Y6 stay below 0.05 under the descriptive Holm guard across the 14 outcome contrasts. Relative to D, C differed only in diff size relative to gold, and that difference does not

survive the guard. D versus A moves in the same direction as C versus A for files touched (rate ratio 0.89, 95% CI 0.82 to 0.97), steps (rate ratio 1.11, 95% CI 1.06 to 1.16) and reaching the step limit (odds ratio 1.61, 95% CI 1.26 to 2.04).

Table S21: Secondary Outcomes: C vs A and C vs D Contrasts (GEE, Clustered by Repository)

			C vs A		C vs D	
Outcome	**Scale**	**Runs**	**Estimate [95% CI]**	p **(Holm g.)**	**Estimate [95% CI]**	p **(Holm g.)**
Y0 patch produced	OR	960	0.58 [0.36, 0.94]	0.027 (0.296)	1.03 [0.68, 1.56]	0.879 (1)
Y1 ΔCCN	diff	855	1.07 [−2.62, 4.77]	0.564 (1)	−1.11 [−5.10, 2.89]	0.582 (1)
Y2 new import edges	RR	855	0.94 [0.67, 1.31]	0.701 (1)	1.01 [0.73, 1.39]	0.974 (1)
Y3 share in stable	OR	726	1.01 [0.95, 1.08]	0.685 (1)	1.00 [0.99, 1.02]	0.592 (1)
Y4 size vs gold	diff	855	−0.66 [−1.06, −0.26]	0.002 (0.023)	−0.42 [−0.80, −0.04]	0.031 (0.310)
Y5 files touched	RR	960	0.86 [0.78, 0.95]	0.005 (0.056)	0.97 [0.89, 1.06]	0.471 (1)
Y6 steps	RR	960	1.09 [1.03, 1.14]	0.001 (0.015)	0.98 [0.94, 1.02]	0.286 (1)
Step limit reached	OR	960	1.37 [1.04, 1.82]	0.027 (n/a)	0.85 [0.65, 1.13]	0.259 (n/a)

OR: odds ratio (Y3 by fractional logit [23]). RR: rate ratio (Poisson). diff: difference in means (Y4 in $\log_2$ units). Holm g.: Holm-adjusted p across the 14 contrasts of Y0 to Y6, a descriptive guard only. All secondary results are hypotheses for future work.

S4.5. Estimators and Power Analysis

The confirmatory GEE [24] uses a binomial family and logit link, clustering by repository, an exchangeable working correlation and the bias-reduced sandwich covariance of Mancl and DeRouen [25]. Wald statistics are referred to $t(60)$ and $F(q, 60)$. Model M1 regresses *resolved* on condition, collapsed stage and model (a fixed effect), and M2 adds the condition by stage interaction. H1 is the C coefficient in M1, H2 is the C minus D contrast in M1, and H3 is the joint test of the C by stage interaction terms in M2. The three p-values are adjusted with Holm's procedure [15] at a familywise level of 0.05. A conditional logistic regression stratified by task and a variational Bayes mixed logit are reported alongside and are never decisive. Risk differences come from a task-level paired bootstrap stratified by declared stage with 10,000 resamples.

The secondary estimators agree with GEE on the signs of H1 and H2. The conditional logit gives odds ratios of 1.10 for C versus A and 1.82 for C versus D ($p = 0.048$). These are conditional on the task, larger than the marginal GEE estimates and not decisive. The variational mixed logit places most of the variance at the task level (standard deviation 2.87 on the logit scale, against 0.19 for the repository).

The power analysis simulated the analysis pipeline on the frozen design, with inputs from a pilot in which 2 of 20 condition A runs were resolved. The pre-registered label is read at a conditional odds ratio of 2.0 for C versus A, a latent intraclass correlation (ICC) of 0.5 and a level of 0.05/3. Power at that point was 0.489, so the design is labelled UNINFORMATIVE. The minimum detectable effect (MDE), the smallest conditional odds ratio on the grid with at least 80% power, is 3.0 (power 0.937). H2 has much less power when D carries part of C's effect, for example 0.396 at an odds ratio of 3.0 for C and its square root for D. Power for H3 was below 0.2 in every simulated cell.

S4.6. Stage-Specific and Within-Stratum Estimates

Table S22 gives the stage-specific odds ratios from model M2. They are descriptive. Within the four declared strata, no C versus A contrast had a confidence interval excluding 1 (odds ratios between 0.88 and 1.46), and the exploratory four-level interaction test gave $p = 0.7538$.

Table S22: Stage-Specific Marginal Odds Ratios From Model M2 (Descriptive)

Collapsed stage	**C vs A [95% CI]**	**C vs D [95% CI]**
ACTIVE (32 tasks)	1.08 [0.69, 1.70]	1.55 [0.85, 2.85]
SERVICING (17 tasks)	1.00 [0.67, 1.50]	1.40 [0.70, 2.80]
PHASE_OUT (31 tasks)	1.05 [0.58, 1.92]	1.18 [0.87, 1.59]

S4.7. Condition B, Uptake and Model Differences

On the 37 tasks where B exists, B runs were resolved less often than A runs (odds ratio 0.75, 95% CI 0.58 to 0.96), and C and B did not differ (odds ratio 1.00, 95% CI 0.60 to 1.66). The B subset is dominated by phase-out tasks, so this comparison is descriptive only. The assistant messages of C runs used a C label term in 3.1% (*diffusiongemma*) and 6.9% (*nemotron-3-super*) of runs, no D run used a D display name, and the phrase *repository context* appeared in at most 9.4% of runs of any condition and model. The exploratory condition by model interaction gave $p = 0.1206$.

S4.8. Further Details Moved From the Main Text

All 1,108 scheduled runs reached a valid end state. Figure 6 shows that C and D runs reached the step limit more often than A runs. Figure S4 and Table S20 give the confirmatory tests. The raw p-value of the C versus D

contrast is 0.0332. It stays above the first Holm threshold of 0.0167 in every sensitivity variant (Table S23).

The odds ratios mean the following. A value of 1 means no difference, and a value of 2 means doubled odds of a fix. The power simulation gave power 0.489 at an odds ratio of 2.0, so the design is labelled UNINFORMATIVE. The minimum detectable effect (MDE) is an odds ratio of 3.0. The plan therefore fixed in advance that a non-rejection means "inconclusive at MDE = 3.0" and never "no effect".

Compared with A, C runs used more steps and made smaller patches, and D runs also used more steps. These differences point to appended context in general, not to the reasoner's labels. They are hypotheses only. Agents did not avoid modules labelled stable (Table S21). The secondary outcomes are described in Section S4.4. Stage-specific estimates are in Table S22.

C and D come from the same pipeline run without any LLM. The manifests were injected into the prompt, because in a feasibility study no agent opened a *CLAUDE.md* file by itself.

S4.9. Sensitivity Analyses

Table S23 summarises SA1 to SA8, and none changes a test result. Apart from the *diffusiongemma*-only fit ($p = 0.743$), the raw H2 p-value stays between 0.030 and 0.124 across variants. It is above the first Holm threshold of 0.0167 in every variant, and grouping by task instead of repository (SA6) gives a confidence interval that includes 1. SA8 keeps only the 31 tasks whose C manifest had no explanation fallback (49 of the 80 manifests had at least one). Its point estimates are larger but its intervals are wide. SA9 recomputes Y3 under seven alternative module-stability thresholds and windows, and no contrast has an interval excluding 1.

S4.10. RQ6: Injected False Claims

This subsection gives the detail behind Sections 4.2.6 and 4.3.5. The run settings are those of RQ4 and RQ5, with the venv restored to a clean copy before every run and before the evaluation of its patch.

Manifests. The true manifest T lists up to 8 layout entries, 3 key files, the test and install commands, up to 5 declared runtime dependencies and the declared Python version. Its median length is 299 characters. Each false item is the first candidate in a fixed order that passes uniqueness constraints: the name occurs nowhere in the base tree or the issue text, so any action on it can

Table S23: Sensitivity Analyses for the Confirmatory Hypotheses

Variant	Tasks	Runs	H1: C vs A OR [95% CI]	H2: C vs D OR [95% CI], p	H3 p
Primary (GEE, by repository)	80	960	1.05 [0.77, 1.42]	1.34 [1.02, 1.74], 0.033	0.967
SA1 exclude infra failures	80	960	1.05 [0.77, 1.42]	1.34 [1.02, 1.74], 0.033	0.967
SA2 *diffusiongemma* only	80	480	0.73 [0.45, 1.20]	1.08 [0.69, 1.67], 0.743	0.207
SA2 *nemotron-3-super* only	80	480	1.29 [0.88, 1.87]	1.51 [1.03, 2.21], 0.035	0.643
SA3 exclude in-tree agent manifest	80	960	1.05 [0.77, 1.42]	1.34 [1.02, 1.74], 0.033	0.967
SA4 length covariate	80	960	1.05 [0.76, 1.44]	1.35 [1.03, 1.76], 0.030	0.967
SA5 exclude gold-drift task	79	948	1.05 [0.77, 1.42]	1.34 [1.02, 1.74], 0.033	0.967
SA6 GEE grouped by task	80	960	1.05 [0.77, 1.43]	1.36 [0.99, 1.86], 0.057	0.967
SA7 conditional logit[a]	80	960	1.10 [0.58, 2.07]	1.82 [1.01, 3.29], 0.048	0.966
SA8 exclude explanation fallback	31	372	1.35 [0.76, 2.37]	2.34 [0.78, 7.05], 0.124	0.775

[a] Odds ratios conditional on the task, so not on the same scale as the marginal GEE estimates.

be attributed to the manifest. Folder names come from a fixed vocabulary (*core*, *internal*, *engine* and others), deleted files from the repository's history, *make* targets from *test*, *tests* and *check* for the test command (F_R) or *install*, *dev* and *develop* for the install command (F_I), and dependencies from a list of competitors such as *httpx* for *requests*. F_R and F_I hold the same number of false claims in every task: 4 in 52 tasks, 3 in 27 and 2 in 1. In F_R, 150 false claims replace a true line and 141 are added. The added lines make the median F_R manifest 13% longer than T. All 1,375 distinct claims passed the checker, the re-check and runtime probes, after one true test-command claim that HermiT contradicted was dropped from its task.

Analysis. Actions are the shell commands the harness executed, and an action is a false lead if its text names a false item as a whole word. A and T runs are scored against the F_R items of their task, which shows how often agents reach the same names without a manifest. Wasted steps add failed commands whose output reports a missing file, command, target or module. The power simulation used the WP5 data and the realised dose, with Holm's first step ($\alpha = 0.0125$). It gave power 0.97 for H1 when each false claim draws an action in 2% of runs, 0.93 for H2 at 0.28 extra wasted steps per run, and 0.90 for H4 at an odds ratio of 0.3.

Table S24 lists the confirmatory and sensitivity estimates. Two estimates go against the hypothesised direction: fewer wasted steps in F_R on the 21 tasks whose false claims all replace a true line, and a higher resolve rate in the tasks where no item needed a fallback placement (raw $p = 0.019$, Holm-adjusted 0.058). Neither is pre-registered as a primary result, and we report them as found. In secondary contrasts, F_I also drew more false-lead actions than T (rate ratio 10.94, 95% CI 1.22 to 98.13), and F_R runs mentioned a

Table S24: RQ6 Confirmatory Hypotheses and Sensitivity Analyses (Rate Ratio or Odds Ratio [95% CI]; n.e. = Not Estimable Because T Has No Action on a False Item)

Analysis	**H1 actions, F_R/T**	**H2 wasted steps**	**H3 actions, F_R/F_I**	**H4 r**
Primary (Holm p)	35.3 [4.1, 300] (0.0045)	1.01 [0.66, 1.53] (0.98)	3.22 [1.65, 6.29] (0.0035)	1.13 [
Clustered by task	34.4 [5.0, 239]	1.01 [0.70, 1.44]	3.21 [1.72, 6.01]	1.14 [
Negative binomial	36.0 [4.3, 301]	1.04 [0.68, 1.59]	3.31 [1.68, 6.53]	
Narrow failure patterns		1.03 [0.66, 1.59]		
Without gold-drift failures	35.1 [4.2, 296]	1.00 [0.65, 1.53]	3.22 [1.65, 6.29]	1.14 [
nemotron only	21.0 [3.2, 137]	1.46 [0.81, 2.62]	2.47 [1.38, 4.43]	1.00 [
diffusiongemma only	n.e.	0.77 [0.47, 1.24]	n.e.	1.35 [
Without generator fallbacks	n.e.	1.13 [0.59, 2.17]	6.75 [1.69, 27.0]	1.43 [
Replaced claims only	n.e.	0.49 [0.29, 0.83]	6.75 [2.30, 19.8]	1.32 [

false item in their reasoning text more often than T runs (rate ratio 3.61, 95% CI 2.17 to 5.99). The undefined test command explains most of the H3 difference (rate ratio 12.67, 95% CI 3.20 to 50.15 for this kind alone), partly because F_I keeps the true test command. The first false-lead action came at a median step of 12.

Shared environment in RQ4 and RQ5. A count over the 1,108 RQ4 and RQ5 trajectories found 12 runs that tried to install a package and 9 that succeeded, in 9 tasks. Up to 49 later runs of those tasks could have used a package that an earlier run installed. The run order within a task was randomised, so the carry-over adds noise rather than favouring a condition.

S4.11. Stronger Agents (Exploratory)

These runs were added after the results above were known, so they are exploratory and carry no Holm claims of their own. RQ6 was repeated with *nvidia/nemotron-3-ultra-550b-a55b* (thinking off), the largest NIM model that returned a well-formed action in a format probe on 2026-10-12, on the same 80 tasks, manifests, conditions, two replicates, agent settings and estimators: 640 runs, all of which reached a valid end state. Leftover background processes from agent commands are killed after every run. Table S25 compares the tallies with the main study.

With the same estimators, nemotron-3-ultra gives H2 rate ratio 3.40 (95% CI 1.91 to 6.05, Holm-adjusted $p = 0.00032$), H3 rate ratio 3.17 (95% CI 1.83 to 5.47, Holm-adjusted $p = 0.00032$) and H4 odds ratio 1.29 (95% CI 0.99 to 1.69, Holm-adjusted $p = 0.11$). The H1 ratio is not estimable

Table S25: RQ6 by Condition: the Two Models of the Main Study (320 Runs per Condition) and nemotron-3-ultra (160 Runs per Condition). A and T Runs Are Scored Against the F_R Items

	Main study (two models)				nemotr	
Condition	**Runs with false lead**	**Actions**	**Wasted steps**	**Resolved**	**Runs with false lead**	**Acti**
A	3	6	149	46	1	
T	1	2	159	43	0	
F_R	59	71	160	48	62	
F_I	19	22	125	43	21	

because T has no action on a false item; the paired task-level Wilcoxon test gives Holm-adjusted $p < 10^{-9}$, with 0.475 more false-lead actions per run in F_R (bootstrap 95% CI 0.356 to 0.600). Of the 76 F_R actions, 54 tried the undefined test command, 10 the false folder, 11 the deleted file and 1 the fake dependency.

RQ4 and RQ5 were repeated with nemotron-3-ultra on all 80 tasks, conditions A, C and D, and B on its 37 tasks, with two replicates: 554 runs, all of which reached a valid end state. NIM stopped serving the model for two days during the runs; runs that failed for that reason or because their environment failed were set aside by end state only and run again from scratch (WP5 deviation 16). We also ran Claude Code (Sonnet 5.5, headless) as the agent, in a container with 4 CPUs, 6 GB of memory and no web tools, with a 30-minute limit and no step limit. In C and D the manifest is the repository's CLAUDE.md file; A has none, and B was not run. It covers the 74 tasks that had every Ultra A, C and D run when it started, with two replicates: 444 runs, all of which reached a valid end state (one run in which Claude Code exited mid-run was run again). Table S26 compares the resolve rates with the main study.

Table S26: RQ4/RQ5 Resolved Runs by Condition: the Two Models of the Main Study, nemotron-3-ultra and Claude Code (Sonnet 5.5)

Condition	**Main study (80 tasks)**	**nemotron-3-ultra (80 tasks)**	**Claude Code (74 tasks)**
A (no manifest)	53/320 (16.6%)	47/160 (29.4%)	74/148 (50.0%)
B (repository files)	18/148 (12.2%)	20/74 (27.0%)	–
C (history-based)	55/320 (17.2%)	45/160 (28.1%)	79/148 (53.4%)
D (flat summary)	43/320 (13.4%)	51/160 (31.9%)	71/148 (48.0%)

With the same estimators, nemotron-3-ultra gives a C versus A odds ratio of 0.94 (95% CI 0.76 to 1.17) and a C versus D odds ratio of 0.84 (95% CI 0.63 to 1.10), and H3 gives $p = 0.56$; the Holm-adjusted p-values are 1.0, 0.59 and 1.0. D versus A gives an odds ratio of 1.13 (95% CI 0.86 to 1.48). A label term from C appeared in 14 of 160 C runs. For Claude Code, with one model, C versus A gives an odds ratio of 1.15 (95% CI 0.91 to 1.45), C versus D 1.25 (95% CI 1.04 to 1.51, raw $p = 0.021$, Holm-adjusted $p = 0.064$), and H3 gives $p = 0.55$. Claude Code needed 5.3 turns per run on average, and a label term from C appeared in 2 of 148 C runs.

S5. Threats to Validity: Protocol Deviations and Power Details

This section lists every deviation from the pre-registered protocols and gives the power details behind the conclusion-validity discussion in Section 4.4. The full deviation logs, with dates and reasons, are part of the replication package.

S5.1. Deviations From the Pre-Registered Protocols

Table S27 summarises the 54 logged deviations (4 for RQ1, 9 for RQ2, 7 for RQ3, 14 for RQ4 and RQ5, and 20 for RQ6); the exploratory replications log 8 more, listed in the replication logs. Three of them touch the evidence most directly. In RQ1, three sampled repositories were run through an earlier pipeline version before the freeze, and their labels were seen (R1). The rules and thresholds had been fixed beforehand and did not change afterwards. In RQ2, the claim normalisation and the manifest parsers were fixed after the first-pass contradicted claims had been inspected (W2). Each fix removes a mechanical false contradiction and none changes a check rule, and both passes are reported in Section 4.3.2. In the agent study, two infrastructure changes were made after the run freeze but before the runs were complete (5.10 and 5.11). The API keys were rotated per call, the worker count was raised from 3 to 4, and three runs that had ended in rate-limit failures were redone. Both decisions were taken from error types and throughput only, and the only outcome information seen was one total count of resolved runs with no breakdown by condition, model, stage or task.

S5.2. Realised Power and Effect Scales in the Agent Study

The variational mixed model put most of the outcome variance at the task level (standard deviation 2.87 on the logit scale, against 0.19 for the

repository). These two standard deviations imply a latent intraclass correlation (ICC) of about 0.7 (our derivation), above the 0.5 assumed by the power label. At an ICC of 0.7, the simulated power at an odds ratio of 2.0 was 0.348, so the realised power may have been lower than the label suggests. The power simulation itself had fitting failures in up to 118 of 1,000 replicates in some H3 cells, which are recorded per cell (deviation 5.9).

The MDE is stated on the conditional (task-specific) odds-ratio scale used in the simulation, whereas the confirmatory GEE estimates are marginal odds ratios. Marginal odds ratios are closer to 1 than conditional ones when tasks differ strongly, so the observed marginal estimates cannot be compared directly with the MDE. The analysis has 61 repository clusters, and the bias-reduced sandwich covariance with $t(60)$ reference distributions keeps the tests close to their nominal level with this number of clusters [25].

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Table S27: Deviations From the Pre-Registered Protocols (Summary)

ID	Deviation	Timing
RQ1 (lifecycle inference)		
R1	Earlier pipeline version run on 3 sample repositories, labels seen, outputs deleted	before freeze, no agreement computed
R2	5 of 12 planned development-set repositories run	before unblinding
R3	Differential check aborted on a repository without LOC series, fixed, analysis re-run with identical primary results	after unblinding, no ground truth used
R4	Stage classes inferred by HermiT (stage B), not by Pellet as the plan stated, definitions unchanged	before freeze
RQ2 (claim verification)		
W1	Tool signature for *xunit.v3* added	after check results seen
W2	Normalisation and parser fixes after inspecting first-pass contradictions, both passes reported	after check results seen
W3	500 missing quotes of one extractor repaired mechanically, strict variant reported	during extraction
W4	Extraction concurrency, reasoning effort and thinking settings changed for latency	before extraction
W5	HermiT-versus-Pellet cross-check of check results impossible (Pellet crash on all 79 manifests)	during verification
W6	Optional slow-model extraction run abandoned after endpoint timeouts	during extraction
W7	T7 not re-checked, T8 and section-category covariate not done	scope
W8	Implementation choices for T6 precision, T4 module matching and drift bisection	during implementation
W9	Sensitivity result directories not committed for size, summary committed	after the runs
RQ4 and RQ5 (agent study, all before the runs were complete unless stated)		
5.1	Condition D widened with further raw monthly counts and extra module rows to reach the length tolerance	before generator freeze
5.2	*RepeatedFormatError* accepted as a valid agent termination	after pilot 1
5.3	Flaky P2P tests removed from a task's list, task dropped only above 10% loss	after pilot 1
5.4	Thinking off, 4,096 tokens, 1,800 s wall time, format-error limit 5	after pilot 1
5.5	Binary file changes dropped from agent diffs	after pilot 2
5.6	Power inputs taken from pilot 2	after pilot 2
5.7	Pellet explanation crashes handled by the pre-specified template fallback	generator freeze
5.8	One unmineable task replaced, B tasks 38 to 37, runs 1,112 to 1,108	before any non-pilot run
5.9	Overflow clamp and per-cell checkpoints in the power runner	before run freeze
5.10	Per-call rotation over four API keys, 4 workers	after run freeze, 32 runs done
5.11	Three final rate-limit failures redone from scratch	after all runs, before analysis
5.12	Gold-drift flags in a separate file, SA9 by direct class	after all runs, before